

A Tale of Two Trajectories

I. Data Preparation & Exploration

The 9 survey answer files (students, faculty, and staff for years 2018, 2021, and 2024) were cleaned and explored in Python. Unnecessary columns were removed including but not limited to 'StartDate', 'EndDate', 'Finished', and all columns with text strings. The column names were changed to the question and all the values were changed to the datatype integer. '-99' was replaced with missing values, assuming these numbers reflect the correspondents not answering the specific question. Averages for each question and all the separate files were calculated and merged split by year. These three average data frames were used to create new Excel documents for easy access and understanding. Empty columns were also removed, and the student files were concatenated into one big file. The outcome was 13 files, now nice and clean.

In addition to the main cleaning of the datasets, the comments sections were extracted from all files. These comments were put into a new dataframe with one column having all the comments and one column having the year the comment was written. The three dataframes with the students' comments were concatenated together, clearly stating the comment and the year. This was also done to the faculty and staff comments. Later exported to excel and comments were categorized using one word for each category. These are the categories for the students' comments: printing, survey, faculty, training, blackboard, WiFi, and library. The comments were

The comments categories for the students were turned into a word cloud using tableau. Only negative comments were used to get a better understanding of what can be improved.

II. Background

This data story was developed in collaboration with the university leadership as part of a broader initiative to assess how well academic support services are meeting the student body's needs in a post-pandemic environment. While the university seeks to serve all community members—students, staff, and faculty—students remain the primary audience for this story. Using data from 2018 (pre-COVID), 2021 (during), and 2024 (post-COVID), the goal is to uncover how the importance, satisfaction, and usage of digital services have shifted and whether past investments—particularly in wireless, ERP systems, and classroom technology—are yielding the intended impact.

During our client consultation, it became clear that strategic insight is more valuable than individual metrics. The leadership is especially interested in trends over time, areas for improvement, and evidence that supports or challenges assumptions—such as the belief that Wi-Fi upgrades would improve satisfaction. Insights from this analysis are intended to support strategic planning efforts including the campus master plan, ongoing digital investments, and the development of outreach and engagement strategies that reflect real community needs.

III. Central Insight

To identify the central insight, the cleaned and aggregated survey data from 2018, 2021, and 2024 was used. Python and pandas were applied for data wrangling and analysis, calculating average Likert-scale responses across three key metrics—importance, satisfaction, and usage—for each academic service. This structure enabled a time-series comparison to observe evolving trends. Year-over-year changes, specifically from the student 2021 to 2024 datasets, were computed to detect increasing or decreasing shifts. Additionally, a 2024-specific gap analysis was conducted to compare importance and satisfaction scores throughout the five-year span, which highlighted discrepancies between expectations and their students’ actual experiences.

This analysis revealed a distinct pattern among students: foundational services such as Wi-Fi, Canvas, and self-service portals, while still performing well, are being increasingly undervalued. For instance, Canvas received a high satisfaction score in 2024 (3.83/4), yet its perceived importance declined to 2.92. Wi-Fi, once a top concern, showed a similar drop in value. The issue is not underperformance, but rather invisibility. As services become more reliable, they fade from conscious recognition, resulting in reduced awareness and appreciation.

In contrast, faculty engagement followed the opposite trajectory. Between 2021 and 2024, “Usage” and “Importance” of classroom technology increased significantly. Usage rose by +0.69, and importance by +0.33. These trends reflect a post-pandemic shift toward deeper integration of digital tools in teaching and learning.

These findings assist the central insight of:

“There is a growing disconnect between the evolving needs and perceptions of students, and the academic services designed to support them.”

There is a disconnect for students when it comes to services that remain important to academic success. For campus leadership, this represents both a risk and an opportunity: without strategic re-engagement, essential student services may become undervalued and underutilized, even as faculty demand for innovation and support continues to grow.

IV. Narrative Development

The narrative, titled “A Tale of Two Trajectories,” was constructed around a central paradox: although students and faculty share similar digital environments, their experiences with and perceptions of academic services diverged in different ways. This insight was connected through a five-part storyline.

The Invisible Backbone

The narrative began in 2018 when services such as Wi-Fi were the most frequently criticized on campus. A significant gap existed between importance and satisfaction, prompting

institutional investment. By 2021, performance and satisfaction improved, and complaints dropped. What trailed, was not increased appreciation, but a notable drop in awareness.

Out of Sight, Out of Mind

By 2024, core services like Wi-Fi and Canvas were operating reliably, yet students no longer ranked them high in importance. Despite Canvas maintaining a strong satisfaction score of 3.83, its importance dropped to 2.92. This suggested that when systems work seamlessly, users stop noticing them—leading to a perception gap that can influence future funding, support, or prioritization.

Meanwhile, in the Classroom...

Unlike the student body, faculty was moving in the opposite direction. Between 2021 and 2024, both the usage and importance of classroom technology increased. Usage rose by +0.69, and importance by +0.33. This trend reflects a turn toward hybrid instruction, remote learning, and larger educational innovation. This momentum was influenced by the pandemic's long-lasting impact and the continued digital transformation of education.

The Growing Disconnect

This divergence in behavior and perception underscores a deeper misalignment. While students are increasingly detached from the very tools that support academic success, faculty are engaging with technology more proactively. The gap is not rooted in functionality, but in perception—and perception directly influences engagement.

Where To Go from Here

The final section of the narrative offers a call to action to realign the campus digital ecosystem. Recommendations include:

- Re-engaging students through campaigns that reconnect them with service value
- Communicating the impact of services, not just maintaining them
- Closing the feedback loop across students, staff, and faculty

This narrative reframes the data into a human-centered insight: if the value of digital tools is not actively surfaced and communicated, those tools will not necessarily fail—but they will fade from visibility, and eventually, from impact.

V. Visualization

To discover fluctuations in the student views of digital services, multiple visualizations were created using cleaned MISO survey data from the years: 2018, 2021, and 2024. The purpose was to highlight the key insight of a growing disconnect: despite high usage and satisfaction, students are increasingly undervaluing these essential services. To maintain continuity in the analysis,

Blackboard and Canvas were combined into a single category: Learning Management System (LMS), reflecting the institutional transition from Blackboard to Canvas between 2021 and 2024.

In the initial visualization, a 2024 side-by-side bar chart, compared usage, importance, and satisfaction scores for the Learning Management Systems (LMS) and Wi-Fi. Although these services received the highest usage and satisfaction scores, their perceived importance was lower. This disconnect served as the base for the central insight.

A subsequent line chart visualized the decline in LMS importance—potentially influenced by the transition between platforms—alongside a rise in classroom technology usage. This highlighted a deviation between valued services and those frequently used. The butterfly chart further visualized the gap between importance and satisfaction, specifically in 2024, pinpointing services where student expectations and experiences were most misaligned.

The time-series line chart shows declining importance scores for Learning Management Systems (LMS), Wi-Fi, and academic tools like library databases. The results suggest that the issue lies not in poor performance but in reduced visibility and perceived relevance. Another chart revealed modest growth in student usage of Ram Search and classroom technology tools from 2021 to 2024, indicating opportunities for increased outreach and student engagement.

The final visualization combined five digital services, showcasing a decline in perceived importance over time. This pattern reinforced the overarching conclusion: effective systems are not enough—perceived value influences student engagement, satisfaction, and institutional investment priorities.

VI. Summary

Through the MISO survey data from 2018, 2021, and 2024, it can be shown a significant change in how students use academic services. While basic services like the Learning Management System (LMS) and Wi-Fi continue to deliver overall high satisfaction and remain heavily used, their perceived importance is declining. Rather than indicating underperformance, this suggests a growing invisibility of digital infrastructure.

The central insight—an emerging disconnect between perception and performance on services for the students—poses both a strategic risk and an opportunity for improvement. Re-engaging students through awareness initiatives and regularly surfacing the value of essential services can help close this perception gap. Without proactive efforts, high-performing systems may continue to fade into the background, ultimately affecting funding, prioritization, and long-term impact. This analysis empowers institutional leadership to prioritize digital service awareness and reinforce perceived value, ensuring continued engagement and impact from mission-critical systems.